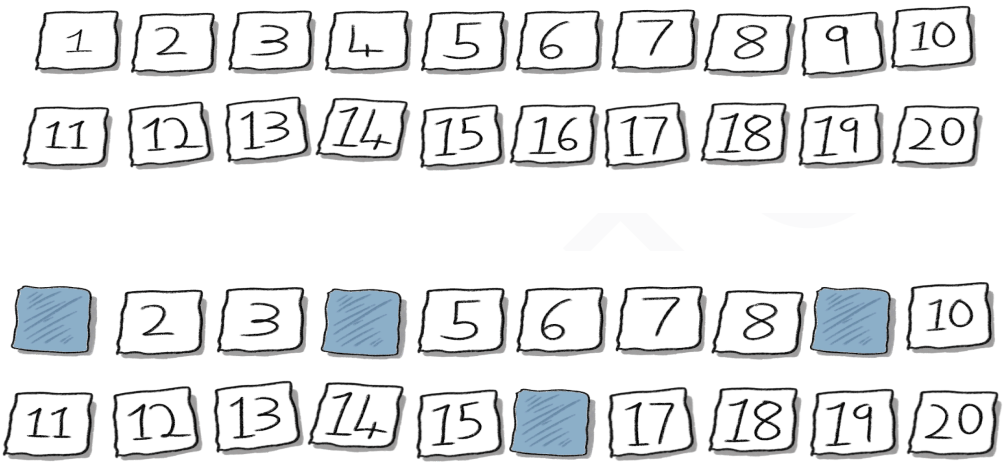
Perfect Squares

Grade 8: A Square and a Cube

Introduction

A regular function of computers is to search for various objects. For this to happen, one needs to devise algorithms cleverly. In this activity, we will explore a method by which we will figure out square numbers from a deck of cards on which consecutive numbers (1-20) are written. By following the simple instructions to flip these cards, we will create an algorithm to find the perfect squares.



This activity helps students understand how to find Perfect Squares (Square Numbers) using their factors.

Activity	Time	Description
Launch	5 min	The teacher demonstrates the activity. Video: https://tinyurl.com/perfectsquarevideo
Trail by students	15 min	Students try out the activity. Worksheet: https://tinyurl.com/perfectsquaresworksheet
Discussions & Explorations	15 min	Attempting the worksheets & the discussion based on the activity.

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CT Components

Algorithmic Thinking: In this activity, students will follow a rule: starting with the first card on which “1” is written, they will flip all cards that are multiples of 1. This becomes the algorithm for every number.

Pattern Recognition: Once students complete flipping cards (1-20), they will try to recognise that some cards’ positions are the same as their starting positions, and some are opposite their starting positions.

Generalisation: After recognising the pattern of card positions relative to their starting positions, students will generalise the positions of cards with **n** positions.

Activity

You are given 20 cards on which consecutive numbers (1-20) are written. You have to flip the cards using a rule.

Rule

Start with “1” and flip all cards that are multiples of 1. Repeat the activity for the number 2, flip all the cards that are multiples of 2. Similarly, repeat the activity for all the numbers from 3 to 20.

What would it look like

1. Starting from “1”, flip all the cards that are multiples of 1.
2. Now, it’s the turn for “2”, flip all the cards that are multiples of 2. Continue doing the same till “10”.
3. From 11 onward, you don’t have multiple selected cards (since we have cards up to 20). So you just flip the selected card.

Q1. Complete the table below:

Cards whose position is the same as the starting	Cards whose position is opposite (w.r.t. starting)

Answer:

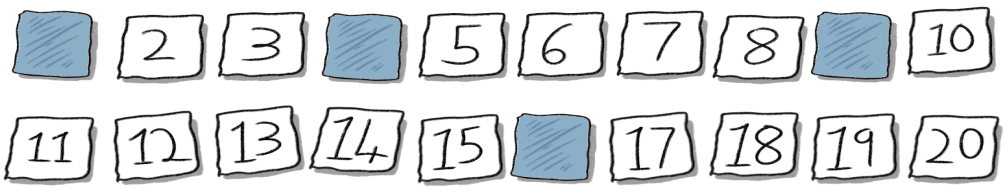
Cards whose face position is the same as the starting	Cards whose face position is opposite (w.r.t. starting)

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2,3,5,6,7,8,10,11,12,13,14,15,17,18, 19,20	**1,4,9,16**

CT Competency: Algorithmic Thinking, Pattern Recognition

Q2. What is special about the numbers in the right column?
A.Prime Numbers
B.Composite Numbers
C.Square Numbers
D.Non square Numbers



Answer: (C) Square Numbers.
CT Competency: Pattern Recognition

How did this happen?

Q3. How many times do you flip the cards on the right column of the table above?

Cards whose face position is the same as the starting	No. of times you flip the card	Cards whose face position is opposite (w.r.t. starting)	No. of times you flip the card
2		1	
3		4	
5		9	
6		16	
7			
8			
10			
11			
12			

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13			
14			
15			
17			
18			
19			
20			

Answer:

Cards whose position is the same as the starting	No. of times you flip the card	Cards whose position is opposite (w.r.t. starting)	No. of times you flip the card
2	2	1	1
3	2	4	3
5	2	9	3
6	4	16	5
7	2		
8	4		
10	4		
11	2		
12	6		
13	2		
14	4		
15	4		
17	2		
18	6		
19	2		
20	6		

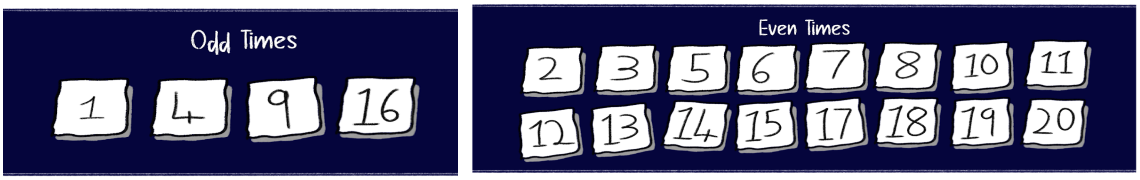
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CT Competency: Pattern Recognition

Q4. In the right column, how many times do you flip the cards?
A. Multiple of 3 times
B.Even times
C.Odd times
D.Multiple of 4 times

Answer: C. Odd times.

CT Competency: Pattern Recognition



Why does this happen? Let's find out.

Q5. Write down the factors of five numbers in the left column. Let's write down the factors of the square number.

Numbers in the left column (any 5 numbers of your choice)	FACTORS (Increasing order)	Square numbers	FACTORS (Increasing order)
		1	
		4	
		9	
		16	

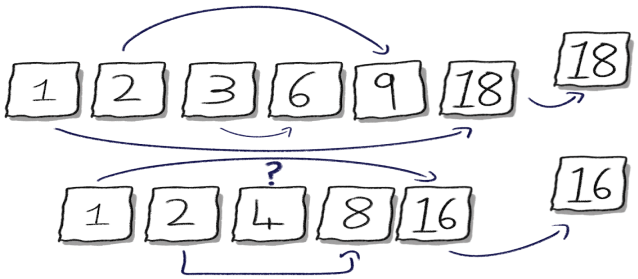
Answer:

Numbers in the left column	FACTORS	Square numbers	FACTORS
3	1,3	1	1
10	1,2,5,10	4	1,2,4
14	1,2,7,14	9	1,3,9
18	1,2,3,6,9,18	16	1,2,4,8,16

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20	1,2,4,5,10,20		

CT Competency: Pattern Recognition, algorithmic thinking



Q6. For the non-square numbers, what do we notice about the 2nd and 2nd-to-last factors, the 3rd and 3rd-to-last factors, etc.?

A.The numbers are not related to each other
B.Their multiplication gives that particular non-square number
C.Their addition gives that particular square number
D.None of the above

Answer:. B. Their multiplication gives that particular non-square number

CT Competency: Pattern Recognition, Generalisation

Conclusion

This activity shows how a simple rule-based process can act like a searching algorithm to identify special numbers. By repeatedly flipping cards, we systematically check positions, much as computers search through data. We discovered that perfect squares behave differently because they have an odd number of factors. This happens because their factors pair up uniquely, with one factor repeated (like $\sqrt{n} \times \sqrt{n}$). Thus, the activity connects algorithms with number properties, helping us understand how factors reveal deeper patterns in numbers.

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